Research Paper Presentation Template

A reusable ISISLab/UNISA Beamer theme for clear scientific storytelling

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Agenda

SECTION 1

Motivation and Problem

Define the research question and why it matters

Problem State the gap, audience, and practical consequence.

One clear research story

Narrative flow

1. Here is the problem
2. It matters to a clear audience
3. There is a gap in current work

Source guidance

A strong research paper centers on one reusable idea and makes the claim explicit. This demo follows the Microsoft Research writing structure: problem, idea, evidence, comparison, conclusion.[a]

[a] Inspired by Simon Peyton Jones, *How to write a great research paper*, Microsoft Research, 2016.

Problem and contributions

Problem

Existing systems are accurate in controlled settings, but performance degrades when the data distribution changes. This creates unreliable outcomes in real deployments.

Research question

Can we design a method that remains accurate, efficient, and interpretable under realistic distribution shifts?

Contributions

1. A new robust learning formulation
2. A reproducible experimental protocol
3. Evidence across three benchmark scenarios

SECTION 02

Related Work and Gap

Position the contribution without turning the talk into a bibliography

Positioning Compare families of approaches around the unresolved gap.

Where the paper fits

12

core papers

3

main families

Comparison logic

Avoid listing related work too early. First explain the problem and the idea; then compare alternatives around the gap the paper addresses.

Gap statement

Prior work optimizes either accuracy or robustness, but does not jointly report reliability, computational cost, and interpretability.

Claims must map to evidence

Claim What we promise → **Method** How we test → **Evidence** Data and analysis → **Interpretation** What it means

Evidence checklist

Every claim introduced in the first part of the presentation should be revisited with a figure, table, proof sketch, or experiment in the results section.

SECTION 03

Method

Move from intuition to a reproducible algorithm

Core idea Explain inputs, transformations, assumptions, and outputs.

Method pipeline

Input Raw data → **Model** Proposed method → **Validation** Metrics → **Release** Code and data

Design principle

Separate the core idea from implementation details. Show the intuition first, then the formal or algorithmic specification.

Method slide rule

One pipeline slide should explain how data becomes evidence. Later slides can zoom into each module.

Algorithmic components

Step 1

Represent each instance with stable and shift-sensitive features.

Step 2

Optimize a robust objective that penalizes unstable shortcuts.

Step 3

Explain decisions with component-level attribution.

Take-home message

The method is designed to preserve performance when the test environment differs from the training environment.

SECTION 04

Experimental Design

Make the evidence credible before showing the results

Protocol Specify data, baselines, metrics, controls, and reproducibility.

Experimental design

Evaluation matrix

1. Three datasets
2. Two distribution shifts
3. Accuracy and robustness metrics
4. Runtime and calibration

Protocol

Use the same train, validation, and test partitions for every baseline. Report average and variance across repeated runs.

3
datasets

5
baselines

Reproducibility package

Data Splits → **Code** Scripts → **Models** Checkpoints → **Report** Figures

Artifact policy

Make the presentation point to the same artifacts as the paper: repository, data card, environment file, and evaluation scripts.

SECTION 05

Results

Lead with the main finding, then test why it holds

Evidence Main result, ablation, uncertainty, and failure cases.

Main result: robustness improves

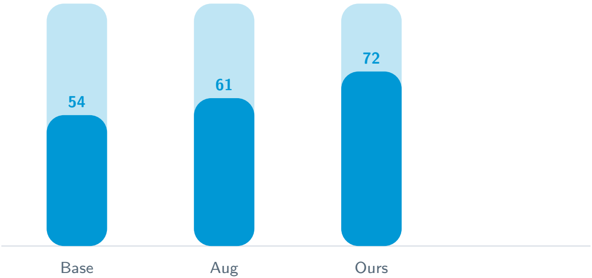
+18%

relative robustness gain

42 ms

median latency

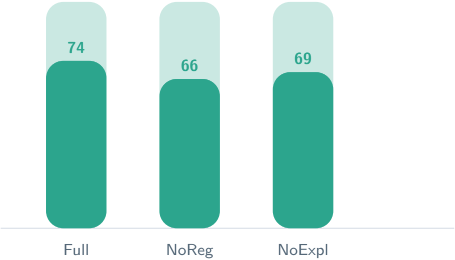
Demo bar chart



Accuracy under shift

Ablation and training dynamics

Ablation



Robust accuracy

Trend

Validation score over epochs

SECTION 06

Discussion and Conclusion

Interpret the evidence and close the research story

Takeaway Limitations, implications, future work, and final message.

Limitations and threats to validity

Limitations

1. Only medium-scale datasets
2. Synthetic shift may not capture all deployment cases
3. Interpretability metric is task dependent

Mitigations

1. Release scripts for replication
2. Include diverse baselines
3. Report failure cases rather than hiding them

Conclusion

Answer to the research question

The method improves robustness under distribution shift while keeping runtime practical and explanations inspectable.

Future work

Scale to larger datasets, validate with field data, and add human-centered evaluation of explanations.

Final message

A great research presentation makes one idea memorable, states refutable contributions, and connects every result back to evidence.

Problem Idea Evidence Impact

SECTION 07

Slide Layout Templates

Copy, replace the images, and keep the same visual grammar

Templates Ready-made layouts for figures, evidence, comparison, and summary slides.

Template: figure left, story right

Hero figure

Replace with the main diagram, model overview, or result figure.

Right-side narrative

1. Name the visual
2. State the key message
3. Explain why it matters

Design cue

Use one large image on the left when the audience should inspect the figure before reading the interpretation.

Template: story left, figure right

Left-side claim

Start with a short claim, then use the figure as evidence. This layout works well for ablations, examples, and qualitative results.

1

claim

1

figure

Supporting figure

Replace with evidence that supports the claim.

Template: two figures comparison

Before / baseline

Describe the reference condition.

After / proposed

Describe the improved condition.

Baseline Proposed Difference

Template: two figures comparison

Before / baseline

Describe the reference condition.

Baseline | Proposed | Difference

Template: four figure gallery

Panel A

Data or task example.

Panel B

Model output or case study.

Panel C

Ablation or variant.

Panel D

Failure case or insight.

Template: figure

Central figure

Use for architecture diagrams, experimental setup, or explanatory graphics.

+18%

gain

42 ms

latency

Interpretation

One sentence connects the figure to the main result. One sentence states the limitation.

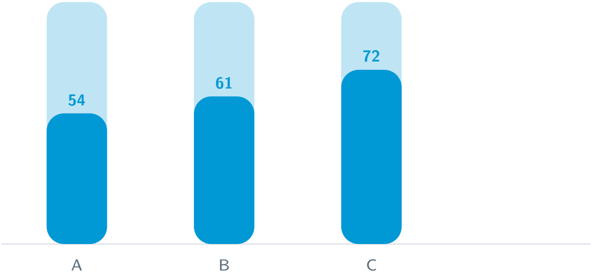
Template: metrics plus chart

74
score

5
baselines

3
datasets

Result chart



Replace labels and values

Main result | Confidence | Caveat

Template: process with output figure

Input Raw source → **Transform** Core method → **Output** Figure or table → **Decision** Action

How to use

Explain the process in the pipeline, then show the concrete output in the neighboring figure.

Output figure

Replace with the artifact produced by the method.

Template: summary and next steps

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Takeaway

The strongest message should fit in two short sentences and repeat the paper's central contribution.

Next step

State the immediate action: reproduce, deploy, validate, extend, or compare.

Style tokens

Problem Method Result Impact

Use badges for status labels, metric cards for numbers, and regular cards for reasoning.

Do not overload

One slide should communicate one job: compare, explain, quantify, or summarize.

References

Cited template source

Simon Peyton Jones. *How to write a great research paper*. Microsoft Research, 2016.
microsoft.com/en-us/research/wp-content/uploads/2016/07/How-to-write-a-great-research-paper.pdf

Questions · Discussion · Next steps